

Scientific gap (1.5)

Research question (1.6)

Chapter and approach (1.7)

1.5.1 Attribution

Canopy structural variables covary in real stands, so a correlative model cannot assign buffering to any one of them.

RQ1

Which dimension of canopy structure governs sub-canopy $\Delta T_{\max}$ buffering: leaf quantity, or vertical arrangement?

Chapter 1

Controlled trait perturbation in MuSiCA around each plot's own measured canopy, over 400 cLHS design plots. ALS only, plot scale.

1.5.2 Inter-sensor consistency

LiDAR and Sentinel-2 both report a leaf area index, but they integrate different parts of the canopy column.

RQ2

Can optical LAI be reconciled with LiDAR-derived structural LAI, or does optical saturation decouple them?

Chapter 2

Sequential correction: effective optical depth, then PROSAIL priors, then horizontal heterogeneity. ALS + S2, three sites, 10 m grid.

1.5.3 Spatio-temporal trade-off

No single sensor delivers three-dimensional structure and temporal dynamics at the same time.

RQ3

Which remotely sensed structure should force the model, and does the answer depend on the canopy archetype?

Chapter 3

Four leaf-area scenarios forced into MuSiCA, scored on two buffering metrics against the sub-canopy plots, read archetype by archetype.

Until we know which structural dimension the microclimate responds to, we cannot say what a remote sensor needs to measure; until we know what part of the canopy the optical sensor sees, we cannot say whether it measures that thing.

the gaps must be closed in this order